

Step 2

Interaction energy:

$$\text{Energy} = -\boldsymbol{\mu} \cdot \mathbf{B}$$

Align spin of ^{60}Co :

$$\frac{n_{\uparrow}}{n_{\downarrow}} = \exp\left(\frac{2\mu B}{k_B T}\right)$$

Bohr and proton
magnetons:

$$\mu_B = \frac{e}{2m_e} \qquad \mu_N = \frac{e}{2m_p}$$

Step 2

$$\mu_B = 5.79 \cdot 10^{-5} \text{ eV/T}$$

$$\mu_N = 3.15 \cdot 10^{-8} \text{ eV/T}$$

$$\frac{\mu_B}{\mu_N} = \frac{m_p}{m_e} \approx 1836$$

$$k_B T = 8.62 \times 10^{-5} \text{ eV/K}$$